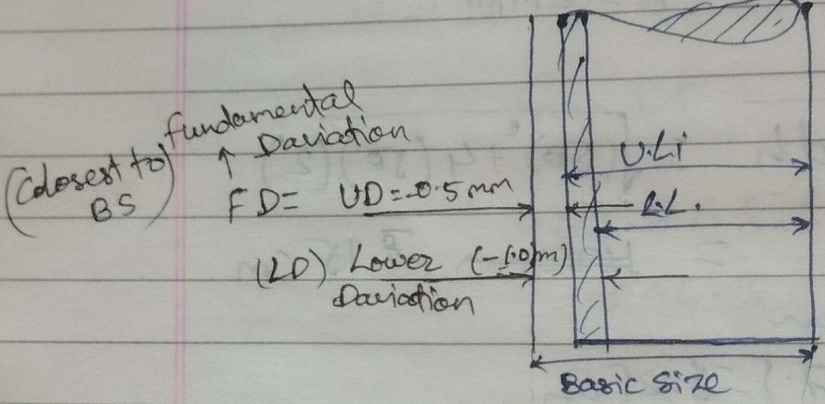
Q Grade of $L_b^{\text{Bend allowance}} = \alpha(R+Kt)$

Limit, Tolerance & Fits

1000 Q shaft
100 mm (B.S) Basic size
(LL) Lower limit = 99 m
(UL) Upper limit = 99.5 m

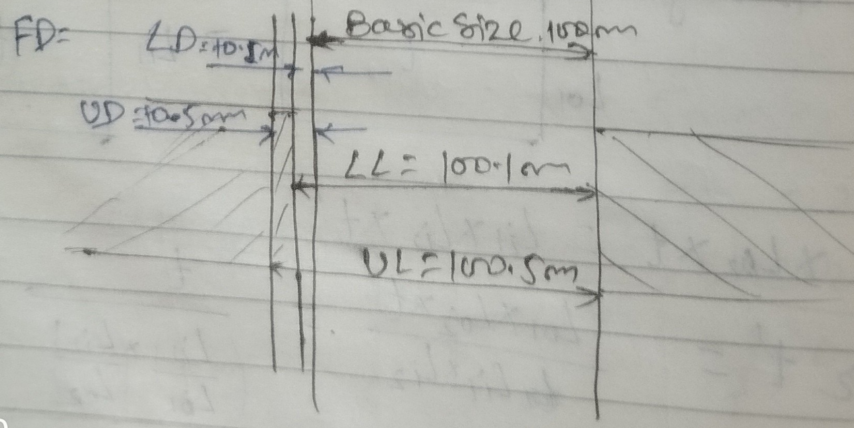


$\boxed{\text{Tolerance} = UL - LL}$

Deviation with sign

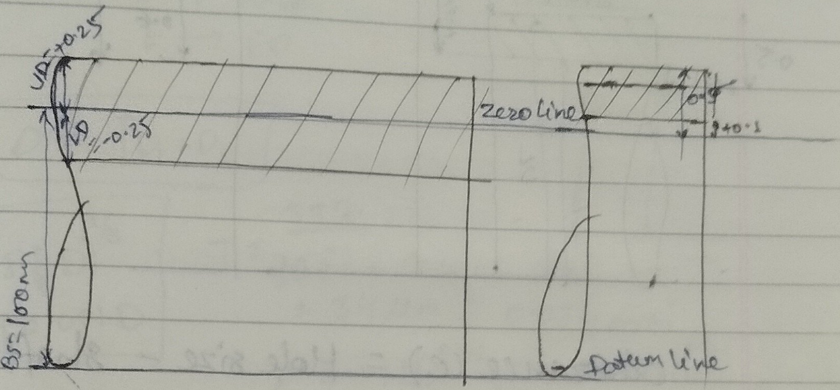
BS_{LL}^{UL} i.e. $\boxed{100^{-0.5}_{-1.0}\ mm}$ shaft

1000 Q
BS = 100 mm
LL = 100.1 mm
UL = 100.5 mm

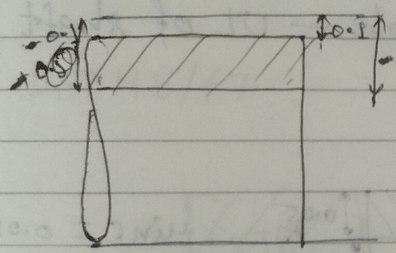


shaft $100^{\pm 0.25}$ mm UD = +0.25 LD = −0.25

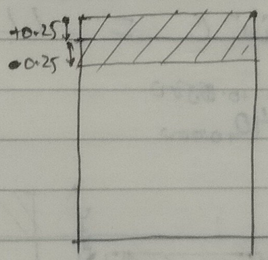
shaft $100^{+0.5}_{+0.1}$



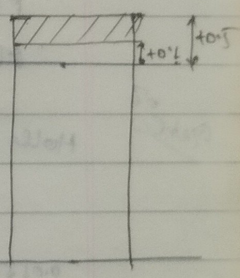
shaft $100^{-0.1}_{-0.5}$



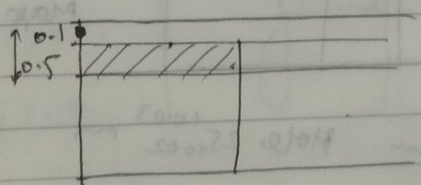
Hole $100^{\pm 0.25}$



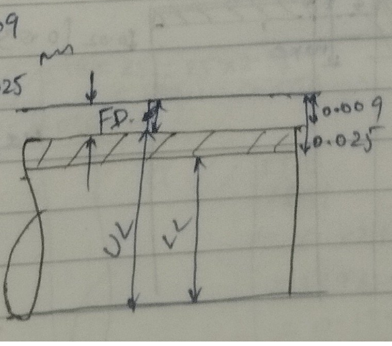
Hole $100^{+0.5}_{+0.1}$



Hole $100^{-0.1}_{-0.5}$



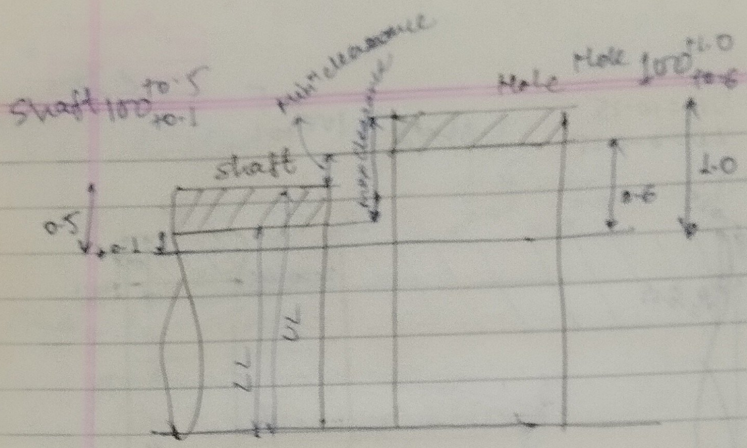
Q Gate shaft $35^{-0.009}_{-0.025}$ mm



Tolerance $T = UL - LL$

$= 0.016$

$FD = -0.009$

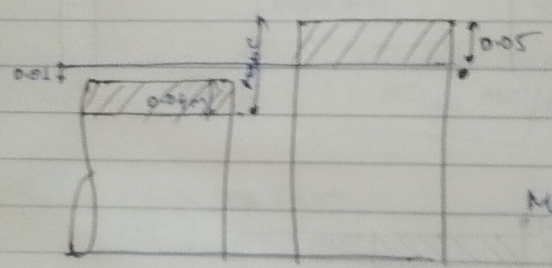


Clearance (C) = Hole size − shaft size

Max^m C = UL of hole − LL of shaft
Min^m C = LL of hole − UL of shaft

Gate 07

Hole $40^{+0.0500}_{+0.0000}$



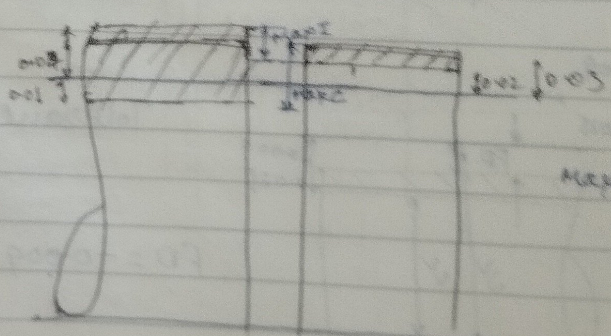
Min C = 0.01 mm

T_{shaft} = 0.04 mm

Max C = 0.04 + 0.01 + 0.05
= 0.10 mm

Gate 012

Shaft $25^{+0.04}_{-0.01}$ mm   Hole $25^{+0.03}_{+0.02}$ mm



Max I = 0.04 − 0.02
= 0.02 mm

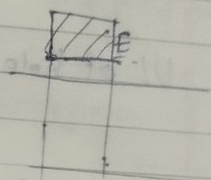
$100\,E7$ ~~IT7~~

BS = 100 mm

E → hole

E → FO

FO = 0 for H



Geometric mean Dia.

$$\boxed{D = \sqrt{D_1 \times D_2}}$$

$$\boxed{i = 0.45\,(D)^{1/3} + 0.001D}$$

$$T = IT7 = 16i$$

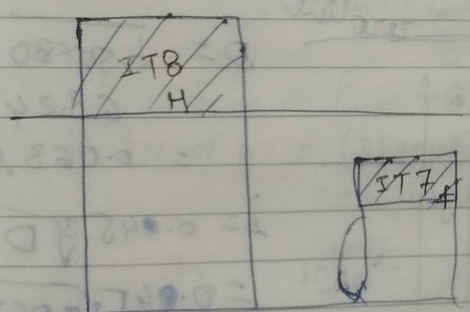
$= 34\,\mu m = 0.034\,mm$

$E = +11D^{0.41}$ $e = -11D^{0.41}$

$= 11 \times 97.98^{0.41}$

$FO = 72\,\mu m$

Q $\phi 50H8/f7$



IES [06]

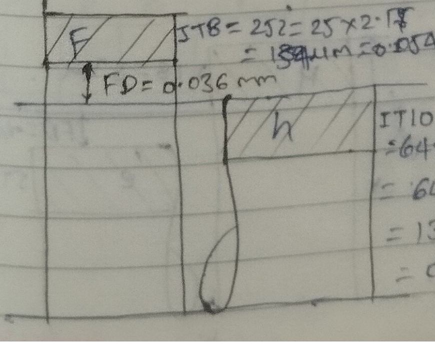
$B = 100\,mm$ F_8h_{10}

$D = \sqrt{80 \times 120} = 97.98\,mm$

$i = 0.45\sqrt[3]{D} + 0.001D$

$= 0.45 \times \sqrt[3]{97.98} + 97.98 \times 0.001$

$= 2.17\,\mu m$



IT8 = 25i = 25 × 2.17

= 54 μm = 0.054 mm

FD = 0.036 mm

IT10

= 64i

= 64 × 2.17

= 139 μm

= 0.139 mm

size

aft

haft

= 0.01 mm

= 0.04 mm

4 + 0.01 + 0.05

0 mm

0.04 − 0.02

0.02 mm

$$F.D.\ of\ F = +5.5D^{0.41}$$
$$= 5.5 \times 97.98^{0.41} = 36\,\mu m = 0.036\,mm$$

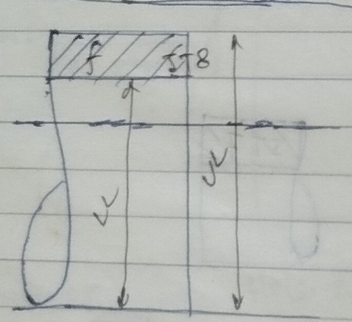
$$UL\ of\ hole = BS + FD + T$$
$$= 100 + 0.036 + 0.054$$
$$= 100.09\,mm$$
$$LL\ of\ hole = 100 + FD = 100.036\,mm$$

$$UL\ of\ shaft = 100\,mm$$
$$LL\ of\ shaft = 100 - 0.139\,\cancel{mm} = 99.861\,mm$$

$$Min^n\ C = 36\,\mu m = Allowance$$
$$Max^m\ C = 54 + 36 + 139 = 219\,\mu m$$

Gate 09

60f8



$$D = \sqrt{50 \times 80}$$
$$= 63.24\,mm$$
$$= 0.063\,\mu m$$
$$i = 0.045\sqrt[3]{D} + 0.001D$$
$$= 0.045\sqrt[3]{0.063} + 0.001 \times 0.063$$
$$= 1.83\,\mu m$$

FD = 30.3 μm

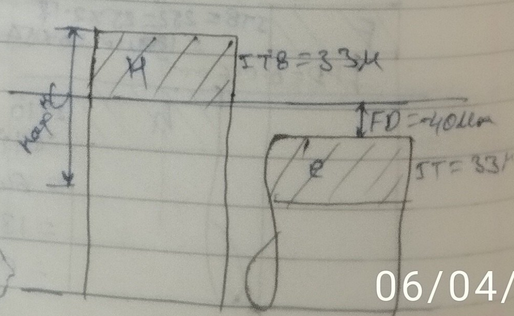
Gate 2000

25H8/e8

$IT8 = 33\,\mu m$

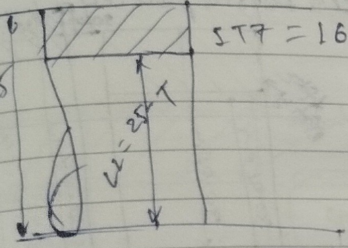
FD for shaft $= -40\,\mu m$

$$Max^m\ C = 33 + 40 + 33$$
$$= 106\,\mu$$



Gate 03

25h7. Taking D = 25 m when diameter is not given



$D = 25\,mm$

$i = 0.45\sqrt[3]{D} + 0.001D$

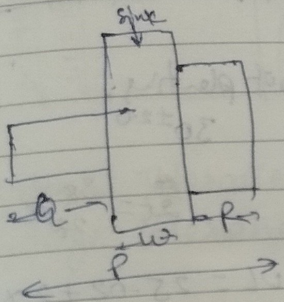
$i = 1.34\,\mu m$

$IT7 = 16i = 16 \times 0.00134$

$= 0.0214\,mm$

$UL = 25.000,\ LL = 24.979\,mm$

Gate 03



$P = 35.00 \pm 0.08$

$Q = 12 \pm 0.02\,m$

$R = 13.00^{+0.04}_{-0.02}$

$P = Q + W + R$

$W = P - Q - R$

$W_{min} = P_{min} - Q_{max} - R_{max}$

$LL\ of\ W = (35-0.08) - (12+0.02) - (13+0.04)$

$= 9.86\,m = 10_{-0.14}$

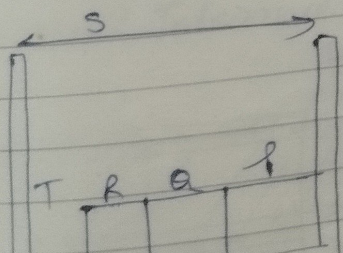
$W_{max} = P_{max} - Q_{min} - R_{min}$

$UL\ of\ W = (35+0.08) - (12-0.02) - (13-0.02)$

$= 10.12\,m = 10^{+0.12}$

$W = 10^{+0.12}_{-0.14} = 9.99 \pm 0.13\,mm$

Gate 97



$S = T + P + Q + R$

$T = S - P - Q - R$

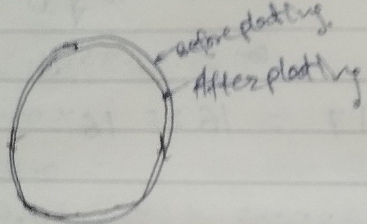
$T_{min} = S_{min} - P_{max} - Q_{max} - R_{max}$

$0.125 = (72.35 + x) - (18.75 + 0.08) - (25 + 0.12) - (28.125 + 0.1)$

$x = -0.05$

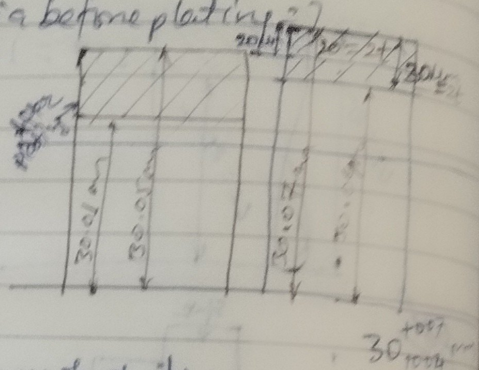
Gate 07

Hole $30^{+0.050}_{+0.010}$ mm after plating.



Thickness $t \to 10-15\,\mu m$

dia before plating = ?

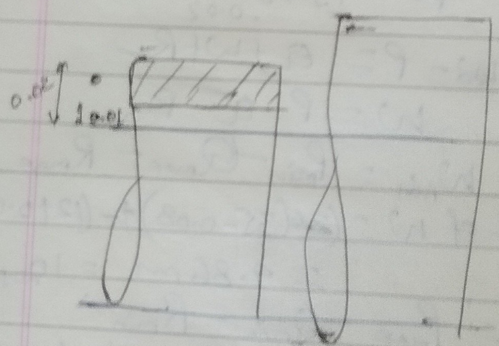


$30^{+0.07}_{+0.04}$ mm

Gate 13

Gauge pin Dia $25^{+0.020}_{+0.010}$

Thickness of plating $30^{\pm 2.0}$



$2t = \frac{32}{28}\,\mu m$

$UL = 25.02 + 2 \times 0.032$

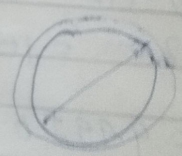
$= 25.02 + 0.064$

$= 25.084$ mm

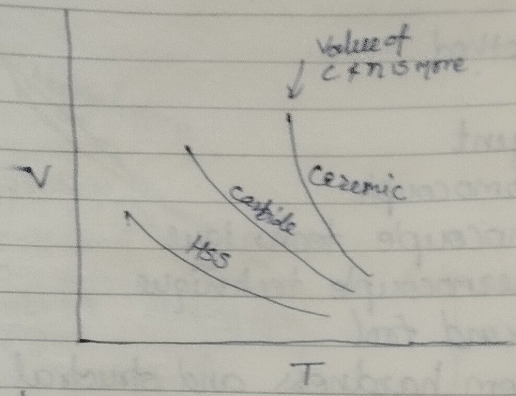
$LL = 25.01 + 2 \times 0.028$

$= 25.01 + 0.036$

$= 25.046$ mm



Tool life curve:-

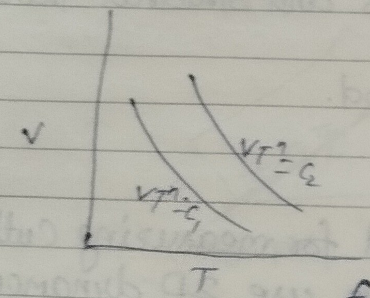


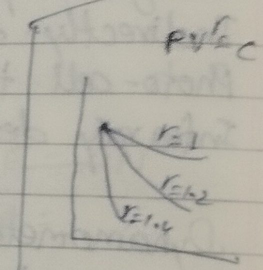
Taylor eq^n

$VT^n = C$

$yx^n = C$

or $y = \frac{C}{x^n}$



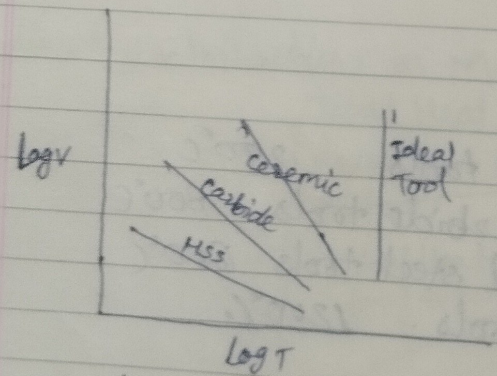


for same material (parallel)

$VT^n = C$

$VT^n = C_1 = 100$

$VT^n = C_2 = 200$

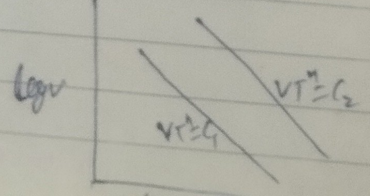


$VT^n = C$

$\log V + n \log T = \log C$

$y + nx = k$

$y = -nx + k$



for same material (‖)

$C_2 > C_1$

Gate 2010

Q. for Tool A $n = 0.45$ $K = 90$

for Tool B $n = 0.3$ $K = 60$

$$x T_A^{0.45} = 90 \Rightarrow T_A = \left(\frac{90}{x}\right)^{\frac{1}{0.45}}$$

$$x T_B^{0.3} = 60 \Rightarrow T_B = \left(\frac{60}{x}\right)^{\frac{1}{0.3}}$$

$$T_A > T_B \qquad \left(\frac{90}{x}\right)^{\frac{1}{0.45}} > \left(\frac{60}{x}\right)^{\frac{1}{0.3}}$$

Or $\left(\frac{90}{x}\right)^{\frac{1}{0.45}} = \left(\frac{60}{x}\right)^{\frac{1}{0.3}}$

$$x = 26.67 \text{ m/min.}$$

Gate 2013

Carbide Tool $VT^{1.6} = 3000$

HSS Tool $VT^{0.6} = 200$

$$\left(\frac{3000}{V}\right)^{\frac{1}{1.6}} > \left(\frac{200}{V}\right)^{\frac{1}{0.6}}$$

$$\frac{1}{1.6} \log\left(\frac{3000}{V}\right) = \frac{1}{0.6} \log\left(\frac{200}{V}\right)$$

$$0.625 \log\left(\frac{3000}{V}\right) = 1.66 \log\left(\frac{200}{V}\right)$$

$$-3.7912 = \cancel{1.66\log} \; 0.625 \cancel{\log V - 1.66} \log V$$

$$= \cancel{-1.035 \log V}$$

$$\log V = 3.6629$$

$$V = \cancel{38.9} \; 39.4 \text{ m/min}$$

~~38.9 m/min~~

Qu: Cutting speed, m/min : $\overset{V_1}{49.74}, \overset{V_2}{49.23}, \overset{V_3}{48.67}, \overset{V_4}{45.76}, \overset{V_5}{42.58}$

Tool life, min : $\underset{T_1}{2.94}, \underset{T_2}{3.90}, \underset{T_3}{\cancel{3.90}}, \underset{T_3}{4.77}, \underset{T_4}{987}, \underset{T_5}{28.27}$

$C = ?$ $VT^n = C$ $\log V + n \log T = \log C$

$x + ny = K$

~~Regr~~ Linear Regration method is used

	V	T	$x = \log V$	$y = \log T$	XY	X^2
1	49.74	$T_1 = 2.94$	$x_1 = \cancel{7.6987}$	$y_1 = 0.4683$	~~1.8301~~	15.272
2	49.23	3.90	$x_2 = 1.6922$	$y_2 = 0.5910$	1.000	
3	48.67	4.77	$x_3 = 1.6872$	$y_3 = 0.6785$	1.1447	
4	45.76	9.87	$x_4 = 1.6604$	$y_4 = 0.9943$	1.6509	
5	42.58	28.27	$x_5 = 1.6260$	$y_5 = 1.4502$	2.3580	

$\Sigma x = 8.36$ $\Sigma y = 4.18$ $\Sigma xy = 6.954$ $\Sigma x^2 = 14$

Gate 03

$y = ax + b$

$y_1 = ax_1 + b$

$y_2 = ax_2 + b$

$y_3 = ax_3 + b$

$y_4 = ax_4 + b$

$y_5 = ax_5 + b$

$\boxed{\Sigma y = a\Sigma x + b \cdot N}$

$y = ax + b$

$\Sigma y = a\Sigma x + b \times 5$

$4.18 = a \times 8.36 + b \times 5$ —— (1)

$xy = ax^2 + bx$ | Multiplying by x both side.

$\Sigma xy = a\Sigma x^2 + b \times \Sigma x$

$6.954 = a \times 14 + b \times 8.36$ —— (2)

By eqn (1) + eqn (2)

$a = -\cancel{7099} = 14.28$

$b = 24.71$

$y = ax + b$

$y = -14.28x + 24.71$

$VT^n = C$

$\log V + n \log T = \log C$

$x + ny = k \log C$

$y = -\frac{1}{n}x + \frac{1}{n}\log C$

$\frac{1}{n} = 14.28$ $\qquad$ $\frac{1}{n}\log C = 24.71$

$n = \frac{1}{14.28} = 0.070$ $\qquad$ $\log C = 1.730$

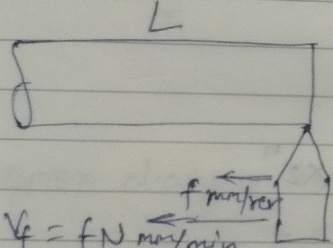
$C = 54$ Ans

Gate 03

$V_1 = \pi D \times 50$ m/min

$V_2 = \pi D \times 80$ m/min

$V_3 = \pi D \times 60$ m/min



10 Tools produce 500 comp
1 Tool - - 50 com

$V_f = fN$ mm/min

$t = \frac{L}{fN}$ min.

$T_1 = 50 \times t_1 = 50 \times \frac{L}{f \times 50} = \frac{L}{f}$ min

$T_2 = 12.2 \times t_2 = 12.2 \times \frac{L}{f \times 80}$

$T_3 = x \times \frac{L}{f \times 60}$

$V_1 T_1^n = V_2 T_2^n = V_3 T_3^n$

$\pi D \times 50 \times \left(\frac{L}{f}\right)^n = (\pi D \times 80) \times \left(12.2 \frac{L}{f \times 80}\right)^n$

or $50 = 80 \times \left(\frac{12.2}{80}\right)^n$

or $\frac{5}{8} = \left(\frac{12.2}{80}\right)^n$ $\Rightarrow n = 0.25$

$$V_1 T_1^{0.25} = V_3 T_3^{0.25}$$

$$\cancel{\pi D} \times 50 \times \left(\frac{L}{\cancel{f}}\right)^{0.25} = \cancel{\pi D} \times 60 \times \left(\frac{x \times \cancel{L}}{\cancel{f} \times 60}\right)^{0.25}$$

or $\frac{50}{6} = 60 \times \left(\frac{x}{60}\right)^{0.25}$

or $\frac{5}{6} = \left(\frac{x}{60}\right)^{0.25}$

or $x = 60 \times \left(\frac{5}{6}\right)^4$

$= 28.935 \simeq 29$

Q IES 06 conv. HSS Tool $T_1 = 1\,hr = 60\,min$

$V_1 = 30\,m/min$

$T_2 = 2.0\,min$

$V_2 = 60\,m/min$

$D = 300\,mm$ for $T_3 = 30\,min$

$V_1 T_1^n = C$ ~~$V_2 T_2^n = V_3 T_3^n$~~

$30 \times 60^n = C$

$60 \times 2^n = C$

$30 \times 60^n = 60 \times 2^n$

$\frac{60}{2} = 2^{1/n}$

$30 = 2^{1/n}$

$\log 30 = \frac{1}{n} \log 2$

$1.477 = \frac{1}{n} \times 0.3010$

$n = 0.2038$

$V_1 T_1^n = V_3 T_3^n$

$30 \times 60^{0.2038} = V_3 \times (30)^{0.2038}$

$V_3 = 35$ m/min

$V_3 = \pi DN \Rightarrow N = \frac{35}{\pi \times 0.3} = 37.13$ rpm.

gate 09 clearance angle changed from 10° to 7° (α).

[Hint] flank wear rate proportion to $\cot\alpha$

rake angle zero tool life = ?

$$\text{Tool life} \propto \frac{1}{\text{Tool wear}}$$

$$T \propto \frac{1}{\cot\alpha}$$

$$T \propto \tan\alpha$$

$$T = k\tan\alpha$$

$$T_1 = k\tan 10$$

$$T_2 = k\tan 7$$

$$\frac{T_2 - T_1}{T_1} \times 100\% = \frac{k\tan 7 - k\tan 10}{\tan 10} \times 100$$

$$= -30\%$$

approx change in tool wear 30% decrease.

IES 09 conventi

$$T_0 = \left(T_c + \frac{C_t}{C_m}\right)\left(\frac{1-n}{n}\right)$$

$T_c = 3$ min $\quad C_t = 3\text{min} \times 0.5 + 5 =$ Rs. 6.5/grind

$C_m =$ Rs 0.50 per min

$n = 0.2$

$$T_0 = \left(3 + \frac{6.5}{0.50}\right)\left(\frac{1-0.2}{0.2}\right) = 64 \text{ min}$$

$$V_0 = \frac{C}{T_0^n} = \frac{60}{64^{0.2}} = 26 \text{ m/min}$$

ESE 01 conv.

$V_1 = 50$ m/min $T_1 = 45$ min

$V_2 = 100$ m/min $T_2 = 10$ min

$V_1 T_1^n = C$

$V_2 T_2^n = C$

$V_1 T_1^n = V_2 T_2^n$

$50 \times 45^n = 100 \times 10^n$

$2 = (4.5)^n$

$\log 2 = n \log 4.5$

$n = 0.46$

$50 \times 45^{0.46} = C$

$C = 288$

Max^n productivity for 2 mins $V_0 = ?$

$T_0 = T_c \left(\frac{1-n}{n}\right)$

$= 2\left(\frac{1-0.46}{0.46}\right)$

$= 2.34$ min.

$V_0 = \frac{C}{T_0^n} = \frac{288}{2.34^{0.46}} = 194$ m/min